

Third Semester
Name of Paper: Operating Systems
Course No.: BCA/3/CC/13

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **4**
(3- 1- 0)

Unit-1: Introduction To Operating System:

Definition of Operating Systems, Functions of Operating Systems, Services of Operating Systems, Types of Operating Systems: Batch, Multiprogrammed, Time sharing, Personal-computer systems, Parallel systems, Real time and Distributed Operating Systems .

Unit-2: Process Management:

Definition of a Process, Process states, Process Control Block, Scheduling Criteria. **Schedulers:** Short term, Medium term and Long term schedulers.

Scheduling Algorithms: FCFS, Round Robin, SJF and Priority Algorithms (preemptive and non-preemptive).

Critical Section Problem: Dining Philosophers, Producer Consumer, Readers Writers–Semaphores – Monitors.

Threads Overview, Multicore Programming, Multithreading models- many-to-one model, one-to-one model, many-to-many model.

Unit-3: Deadlock:

Definition of a Deadlock, Deadlock Characterization: Necessary Conditions, Resource-Allocation Graph, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance-Safe State, Resource-allocation-Graph Algorithm-Banker's Algorithm. Deadlock Detection, Recovery from Deadlock.

Unit-4: Memory Management:

Memory Management: Background, Logical versus Physical Address space, swapping, Contiguous allocation, Paging, Segmentation. **Virtual Memory:** Demand Paging, **Page Replacement:** FIFO and LRU

Unit-5: File System:

File Concept, File Attributes, File Operation, File Types. Access Methods-Sequential Access and Direct Access. Directory Structure-Single-Level Directory, Two-Level Directory, Tree-Structured Directories, Acyclic-Graph Directories.

File protection- Access Control List (ACL), other protection approaches, Allocation methods: Contiguous, linked and index allocation.

Reference Books:

1. Silberschatz, Abraham and Galvin, Peter Baer: *Operating System Concepts*. 9th Edition. John Wiley and Sons
2. Milan Milenkovic: *Operating Systems Concepts and Design*, McGraw Hill 1987
3. Tanenbaum, S Andrew: *Operating Systems – Design and Implementation*, PHI, 2001.
4. Stalling Williams: *Operating Systems*, PHI, 2001

Third Semester
Name of Paper: Data Structure using C
Course No.: BCA/3/CC/14

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **4**
(3- 1- 0)

Unit-1: Introduction to Data Structures

Definition- Classification of data structures (Linear and Non-Linear), Operations on data structures. Complexity - Time-Space complexity.

Review of Arrays, Structures and Unions. Pointers and Dynamic Memory Allocation Functions- Malloc, Calloc, free and reallocation function.

Unit-2: Stack & Queues

Stacks: Concepts, Operations, Application of stacks- conversion from infix to postfix and prefix expression, evaluation of postfix expression using stacks (no algorithms), Recursion – Factorial, GCD, Fibonacci Sequence, Tower of Hanoi

Queues: Concepts, operations, Linear Queue (FIFO), Circular queues, and Deque (Introductory concepts)- Application of queues.

Unit-3: Searching and Sorting

Searching: Sequential and Binary Search.

Sorting: Bubble sort, Selection sort, Insertion sort, Merge sort, Quick sort.

Unit-4: Linked Lists

Linked list: Representation of linked lists in memory, Operations on linked list (Insertion, Deletion, Display), Circularly linked linear lists (Insertion, Deletion, Display), Doubly linked linear list (Insertion, Deletion, Display), Applications of linked linear lists.

Unit 5: Graphs and Trees

Trees: Binary trees, Complete Binary tree, Binary Search Trees – Searching and Inserting in Binary Search Trees, Deletion in a binary search tree. Traversals on a Binary Tree (In-order, post-order, pre-order), Application of Trees.

Graphs: Traversal methods: Breadth First Search and Depth First Search.

Recommended Books:

1. A.M.Padma Reddy : *Systematic Approach to Data Structures using C*, United Publishers (2012)
2. Seymour Lipschutz: *Data Structures*, Schaum's Outlines, Revised 1st Edition, McGraw Hill, 2014
3. Gilberg & Forouzan: *Data Structures: A Pseudo-code approach with C*, 2nd Edition, Cengage Learning, 2014.

Third Semester
Name of Paper: Database Management Systems
Course No.: BCA/3/CC/15

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **4**
(3- 1- 0)

Unit-1: Introduction, Database Architecture and Modeling

(10 L)

Introduction to Database: Importance of database, Characteristics of database approach, difference between database approach and File oriented approach, Characteristics of data in database, meaning of database system, Database users; Database management systems (DBMS): Meaning, advantages of using a DBMS; **Types of DBMS:** Hierarchical, network, relational, object-oriented, object-relational; **Database system concepts and architecture:** Data Models, schemas, and instances, Three schema architecture, Data Independence; Design Constraints.

Unit-2: Entity Relationship Model and Relational DBMS:

(10 L)

Conceptual database model: ER Model, Defining relations; **Components of ER model:** Entities and attributes, Entity types, entity sets, keys and value set, symbols of ER-Diagram; **Relationship:** one-to-one, one-to-many, many-to-many, examples of ER-Diagram; **Enhanced Entity-Relationship (EER):** Super classes, subclasses and inheritance, Specialization and Generalization and their constraints. **Introduction to RDBMS Terminologies:** characteristics of relation, Domain constraints, Entity, attributes and tuples, relational data integrity.

Unit-3: Data normalization, Relational Algebra & Relational Calculus:

(10 L)

Type of Key attributes; Functional dependencies; **Introduction to Normalization:** General definitions and its importance; **Normal forms:** First, second, third, fourth, fifth form, Boyce-Codd Normal form. **Relational algebra:** Relational algebraic operations - (union, intersection, difference, project, Cartesian product, rename, select, division, join.), Examples of queries in Relational Algebra; **Relational Calculus:** Tuple relational calculus, Domain relational calculus.

Unit-4: Introduction to SQL.

(10 L)

Characteristics and advantage of SQL, **Data types:** data types, literals, string, numeric; specifying constraints in SQL. **Types of SQL commands:** DDL, DML, DCL; **SQL operations:** arithmetic, comparison, logical and set operators, Operator precedence, Examples of basic Queries and sub queries. **Aggregate functions:** Applications, general rules, examples of Aggregate functions provided by SQL. **Table:** create, modify, alter, drop. **Views:** and indexes. Insert, update and Delete operations. Joins, unions, Intersections, Minus. Cursors in SQL, Embedded SQL.

Unit-5: Back, Recovery, Database Security and Integrity

(10 L)

Database Security: Requirement of Database security, Dimension, risks and security requirements, Giving Privileges to protecting the data, Authorization and Views, data encryption.

Database backups: Causes of failures, Importance of backup, Back-up mechanism logging, checkpointing. Recovery techniques: Deferred update, intermediate update, shadow paging; Detached transaction actions; Recovery in multi-database systems.

Integrity constraints: basic concepts, Types, Restriction on integrity constraints, Data security risk. Granting and revoking privileges and roles.

Recommended Books:

1. A. Leon & M. Leon: *Database Management Systems*, Vikas Publication House Pvt.Ltd (2008).
2. R. Elmasri, S Navathe: *Fundamentals of Database System*, Pearson / Addison Wesley; 5th edition (2006).

Third Semester
Name of Paper: Computer Organization and Architecture
Course No.: BCA/3/CC/16

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **5**
(4- 1- 0)

Unit-1: **(10 L)**

Register Transfer and Micro Operations: Register Transfer Language, Register Transfer, Bus and Memory Transfer – three state bus buffer, memory transfer, binary adder, binary adder subtractor, binary incrementer, arithmetic circuit, Logic Micro Operations, Shift Micro Operation, Arithmetic Logic Shift Unit.

Unit-2: **(10 L)**

Basic Computer Organization and Design: Instruction Codes – Stored Program Organization, Indirect Address; Computer Registers; Computer Instructions – Instruction Set Completeness, Timing and Control; Instruction Cycle – Fetch and Decode; Input- Output and Interrupt – Input-Output configuration, Input-Output Instructions; Design of Basic Computer.

Unit-3: **(10 L)**

Central Processing Unit: Introduction, General Register Organization – Control Word, Examples of Microoperations; Stack Organization – Register Stack, Memory Stack; Instruction Formats – Three-Address Instructions, Two-Address Instructions, One-Address Instructions, Zero-Address Instructions, RISC Instructions; Addressing Modes – Different types of Addressing Modes; Data Transfer and Manipulation – Data Transfer Instructions, Data Manipulation Instructions, Arithmetic Instructions, Logical and Bit Manipulation Instructions, Shift Instructions; Program Control; Reduced Instruction Set Computer- CISC Characteristics, RISC Characteristics.

Unit-4: **(10 L)**

Input-Output Organization: Peripheral devices; Input-Output Interface - I/O Bus and Interface Modules, I/O versus Memory bus, Isolated versus Memory-Mapped I/O, Example of I/O interface; Asynchronous data transfer – Strobe control, Handshaking; Modes of transfer – Programmed I/O, Interrupt-Initiated I/O; Priority Interrupt; Direct Memory Access - DMA Controller, DMA Transfer; Input-Output Processor – CPU-IOP Communication.

Unit-5: **(10 L)**

Memory Organization: Memory Hierarchy; Main Memory - RAM and ROM chips, Memory Address Map, Memory Connections to CPU; Auxiliary Memory- Magnetic Disk; Cache Memory - Associative Mapping, Direct Mapping, Set-Associative Mapping, Writing into Cache, Cache Initialization; Virtual Memory- Address and Memory Space, Address Mapping Using Pages, Associative Memory Page Table, Page Replacement.

Recommended Books:

1. M.Morris Mano: *Computer System Architecture*, 3rd Edition, Pearson Education, New Delhi (2006).
2. W. Stallings: *Computer Organization & Architecture*, 9th Edition, Pearson Education, New Delhi (2012).

Third Semester
Name of Paper: Oracle Lab
Course No.: BCA/3/CC/17

Marks Scale: **100** marks (End Sem. Exam: 75+Internal: 25)

Credit: **3**
(0- 0- 3)

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|-----------|--------------------|-------------------|
| A. | Record Book | (10 marks) |
| B. | Viva Voce | (15 marks) |
| C. | Practical | (50 marks) |

List of Practical:

1. Create following Three Tables

I. Salesman

SNUM	SNAME	CITY	COMMISSION
1001	PIYUSH	LONDON	12%
1002	NIRAJ	SURAT	13%
1003	MITI	LONDON	11%
1004	RAJESH	BARODA	15%
1005	ANAND	NEW DELHI	10%
1006	RAM	PATAN	10%
1007	LAXMAN	BOMBAY	09%

SNUM : A Unique number assign to each salesman.

SNAME : The name of salesman.

CITY : The location of salesman.

COMMISSION: The salesman commission on order.

II. Customer

CNUM	CNAME	CITY	RATING	SNUM
2001	HARDIK	LONDON	100	1001
2002	GITA	ROME	200	1003
2003	LAXIT	SURAT	200	1002
2004	GOVIND	BOMBAY	300	1002
2005	CHANDU	LONDON	100	1001
2006	CHAMPAK	SURAT	300	1007
2007	PRATIK	ROME	100	1004

CNUM : A Unique number assign to each customer.

CNAME : The name of customer.

CITY : The location of customer.

RATING : A level of preference indicator given to this customer.

SNUM : A salesman number assign to this customer.

III. Orders

ONUM	AMOUNT	ODATE	CNUM	SNUM
3001	18.69	10/03/99	2008	1007
3002	767.19	10/03/99	2001	1001
3003	1900.10	10/03/99	2007	1004
3004	5160.45	10/03/99	2003	1002
3005	1098.25	10/04/99	2008	1007

3006	1713.12	10/04/99	2002	1003
3007	75.75	10/05/99	2004	1002
3008	4723.00	10/05/99	2006	1001
3009	1309.95	10/05/99	2004	1002
3010	9898.87	10/06/99	2006	1001

1. List of all orders for more than Rs. 1000
2. List all customers whose name begins with a letter 'C'
3. Count all Orders of 10th March 1999
4. List all customers serviced by salesman with commission above 12%.
5. Produce the name and rating of all customers who have above average orders.
6. Double the commission of all salesmen of London
7. Calculate the total of orders for each day.
8. Create a view called Big orders which stores all orders larger than Rs. 4000.
9. Create a view that shows all the customers who have the highest ratings.
10. Remove all orders of customer Chandu from the Orders table

2. Consider the Insurance database given below. The primary keys are underlined and the data types are specified:

PERSON	(<u>driver-id</u> :string, name:string, address:string)
CAR	(<u>Regno</u> :string, model:string, year:int)
ACCIDENT	(<u>report-number</u> :int, date:date, location:string)
OWNS	(<u>driver-id</u> :string, regno:string)
PARTICIPATED	(<u>driver-id</u> :string, <u>regno</u> :string, <u>report-number</u> :int, damage-amount:int)

1. Create the above tables by properly specifying the primary keys and the foreign keys
2. Enter at least five tuples for each relation
3. Demonstrate how you
 - a) update the damage amount for the car with a specific regno in accident with
 - b) report number 12 to 25000
 - c) add a new accident to the database
4. Calculate the total damage amount
5. Find the lowest and highest amount of money spent in accident.
6. Select location of accident using subquery.
7. Find the total number of people who owned cars that were involved in accidents in 2002.
8. Find the number of accidents in which cars belonging to a specific model were involved.
9. Create a VIEW called OWNERSHIP that will appear the following :
Driver name, address, registration and location
10. Remove the model column from the CAR table.

3. Consider the following relations for an order processing database applications in a Company

CUSTOMER	(<u>cust</u> :int, cname:string, city:string)
ORDER	(<u>order</u> :int, odate:date, cust:int, ord-amt:int)
ORDER_ITEM	(<u>order</u> :int, <u>item</u> :int,qty:int)
ITEM	(<u>item</u> :int,unitprice:int)
SHIPMENT	(<u>order</u> :int, <u>warehouse</u> :int,ship-date:date)
WAREHOUSE	(<u>warehouse</u> :int,city:string)

- 1 Create the above tables by properly specifying the primary keys and the foreign keys
- 2 Enter at least five tuples for each relation
- 3 Produce a listing: CUSTNAME, # of orders, AVG_ORDER_AMT, where the middle column is the total no of orders by the customer and the last column is the average order amount for that customer
- 4 List the order # for orders that were shipped from all warehouses that the company has in a specified city
- 5 Demonstrate how you delete item #10 from ITEM table and make the field null in the ORDER_ITEM table.
- 6 List the orders date, items and unit price.
- 7 Calculate the total of orders for each day.
- 8 Find out which unit price is lowest.
- 9 Create a VIEW called Big which show all orders larger than ₹2000
- 10 Select unit price in order processing using subquery

4. Consider the following database of student enrollment in courses and books adopted for each course-

STUDENT	(<u>regno</u> :string, name:string, major:string, bdate:date)
COURSE	(<u>course</u> :int, cname:string, dept:string)
ENROLL	(<u>regno</u> :string, <u>course</u> :int, marks:int)
BOOK_ADOPTION	(<u>course</u> :int, sem:int, book-ISBN:int)
TEXT	(<u>book-ISBN</u> :int, book-title:string, publisher:string, author:string)

1. Create the above tables by properly specifying the primary keys and foreign keys
2. Enter five tuples for each relation
3. Demonstrate how you add a new text book to the database and make this book be adopted by some department
4. Produce a list of text books in alphabetical order for courses offered by BCA department that use more than two books
5. List any department that has all its adopted books published by a specific publisher
6. Select marks of the student using sub query.
7. List out student marks in ascending order
8. Add new column position in enroll table
9. Create a view Black Market that gives the count of no. of publisher.
10. Delete the student bdate from the student table.

5. The following tables are maintained by a book dealer-

AUTHOR	(<u>author-id</u> :int, name:string, city:string, country:string)
PUBLISHER	(<u>publisher-id</u> :int, name:string, city:string, country:string)
CATALOG	(<u>book-id</u> :int, title:string, author-id:int, publisher-id:int, category-id:int, year:int, price:int)
CATEGORY	(<u>category-id</u> :int, description:script)
ORDER-DETAILS	(<u>order-no</u> :int, <u>book-id</u> :int, quantity:int)

1. Create the above details by properly specifying the primary keys and foreign keys
2. Enter at least five tuples for each relation
3. Find the author of the book which has maximum sales
4. Demonstrate how you increase the price of books published by a specific publisher by 10%
5. List all authors whose name begins with a letter 'L'.
6. Select the price and author using subquery.
7. Select the order detail ordered by quantity.
8. Shows the total and average quantity of book order
9. Create a view called Booking which shows author name, book id, price, and year
10. Delete the quantity of book orders.

6. Consider the following database for a banking enterprise

BRANCH	(<u>branch-name</u> :string, branch-city:string, assets:real)
ACCOUNT	(<u>accno</u> :int, branch-name:string, balance:real)
DEPOSITOR	(<u>customer-name</u> :string, <u>accno</u> :int)
CUSTOMER	(<u>customer-name</u> :string, customer-street:string, city:string)
LOAN	(<u>loan-number</u> :int, branch-name:string, loan-number-int)
BORROWER	(<u>customer-name</u> :string, customer-street:string, city:string)

1. Create the above tables by properly specifying the primary and foreign keys
2. Enter 5 tuples for each relation
3. Find all the customers who have at least two accounts at the main branch
4. Find all the customers who have an account at all the branches located in a specified city
5. Find the average loan taken by each customer in any branch
6. Select the borrower name and balance using sub query.
7. Find the lowest and highest balance in account table.
8. Find the customer Sanjit take loan from the Noida branch.
9. Create a view called Personal loan that shows customer name, account no and loan
10. demonstrate how you delete all account tuples at every branch located in a specified city

Note: The above lists of experiments are suggestive of the standard. Department may modify, add or delete any experiment whenever it considered necessary

Third Semester
Name of Paper: Data Structure using C Lab
Course No.: BCA/3/CC/18

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **3**
(0- 0- 3)

A. Record Book

(10 marks)

B. Viva Voce

(15 marks)

C. Practical

(50 marks)

1. Write a C Program to create a Sequential file with at least 4 records, each record having the shown below.

Roll No : Non zero positive integer.

Student Name : Character string of length 25

Marks 1, Marks 2, Marks 3: Positive Integer

2. Write a C Program to convert and print a given valid parenthesized infix arithmetic expression to postfix expression. The expression consist of single character operands and the binary operator +, -, * and /.

3. Write a C Program to construct a Stack of integers and to perform the following operations on it.

1. Push

2. Pop

3. Display

The program should print appropriate messages for stack overflow, stack underflow and stack empty

4. Write a C Program to simulate the working of a queue of integers using an array. Provide the following operations.

1. Insert

2. Delete

3. Display

5. Write a C Program to simulate the working of a circular queue of integers using an array. provide the following operations.

1. Insert

2. Delete

3. Display

6. Write a C program to implement linear search.

7. Write a recursive C Program for

1. Searching an element on a given list of integers using the Binary Search method.

2. Solving the Tower of Hanoi problem Q9. Write a C program using Selection Sort

8. Write a C program using Merge Sort

9. Write a C program using bubble Sort

10. Write a C program using insertion Sort

11. Write a C program using quick Sort

12. Write a C Program to construct singly linked list and perform the following operations:

1. insert(first, middle, last position)

2. delete(first, middle, last position)

3. Display

13. Write a C Program to construct doubly linked list and perform the following operations:
1. insert(first, middle, last position)
 2. delete(first, middle, last position)
 3. Display

14. Write a C Program to construct circular linked list
1. insert
 2. delete)
 3. Display

15. Write a C Program using dynamic variables and pointers to construct a stack of integer using linked list and to perform the following operations:

1. Push
2. Pop
3. Display

The program should print appropriate messages for stack overflow, stack underflow and stack empty

16. Write a C Program
1. To construct a binary search tree of integers.
 2. To traverse the tree using all the methods i.e, inorder, preorder and postorder
 3. To display the elements in the tree.
17. Write a recursive C Program for
1. Searching an element on a given list of integers using the Binary Search method.
 2. Solving the Tower of Hanoi problem

Note: *The above lists of experiments are suggestive of the standard. Department may modify, add or delete any experiment whenever it considered necessary*